

MATH 3260 - Chapter 6 Worksheets

Linear Transformations

Name: _____

Date: _____

These are student practice worksheets. They are designed to be used with the chapter notebook and notes, but every problem below contains the data needed to work it independently. No solutions are included in this student version.

Section 6.1 - Introduction to Linear Transformations

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. For $T(x, y) = (2x - y, x + 3y)$, find $T(4, -2)$.
2. Find the preimage of $(5, 6)$ under the transformation in Problem 1.
3. Verify both linearity conditions for $T(x, y) = (x - 2y, 3x + y)$.
4. Explain why $F(x, y) = (x + 1, y)$ is not linear. Identify a specific linearity condition that fails.
5. Suppose

$$T(\mathbf{e}_1) = (9, 1), \quad T(\mathbf{e}_2) = (20, -3), \quad T(\mathbf{e}_3) = (21, 4).$$

Use only these basis images and linearity to find $T(1, -2, 3)$.

6. Consider differentiation $D : P_3 \rightarrow P_2$ and integration $I : P_1 \rightarrow \mathbb{R}$ defined by $I(p) = \int_0^1 p(x) dx$. Find

$$D(3 - 2x + 5x^2 - x^3)$$

and

$$I(1 + x).$$

State the domain and codomain of each transformation.

Section 6.2 - Kernel and Range

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. For $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$T(x, y, z) = (x, y, 0),$$

find $\ker(T)$ and $\text{Range}(T)$. Explicitly state that the kernel is a subspace of the domain and the range is a subspace of the codomain.

2. For

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix},$$

view A as a map $\mathbb{R}^4 \rightarrow \mathbb{R}^3$. Find a basis for $\ker(A)$ and find $\text{rank}(A)$.

3. Verify rank-nullity for the matrix in Problem 2.
4. Determine whether the transformation in Problem 2 is one-to-one, onto, both, or neither. Justify using kernel and rank information.
5. Show that $C(a + bx + cx^2) = (a, b, c)$ is an isomorphism from P_2 to $\mathbb{R}^3$.
6. Explain why the kernel of every linear transformation contains the zero vector.

Section 6.3 - Matrices for Linear Transformations

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Find the standard matrix for

$$T(x, y, z) = (x + 2y - z, 3x - y + 4z).$$

Show how each column comes from $T(\mathbf{e}_j)$.

2. Use the matrix from Problem 1 to find $T(1, 2, -1)$. Then compute $T(1, 2, -1)$ directly from the formula and verify that the two methods agree.

3. Let

$$A_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Find A_2A_1 and A_1A_2 . State which matrix represents “apply A_1 first, then A_2 .”

4. Find the inverse transformation for

$$T(\mathbf{x}) = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \mathbf{x}.$$

Write your final result as $T^{-1}(\mathbf{y}) = A^{-1}\mathbf{y}$.

5. Find the matrix of $T(x, y) = (x + y, 2x - y)$ relative to the ordered bases

$$B = ((1, 1), (1, -1)), \quad C = ((1, 0), (1, 1)).$$

Show explicitly that column j is $[T(\mathbf{b}_j)]_C$.

6. Write the matrix of differentiation $D : P_3 \rightarrow P_2$ relative to the standard polynomial bases.

Section 6.4 - Transition Matrices and Similarity

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. For the ordered bases

$$B = ((1, 1), (1, -1)), \quad C = ((1, 0), (1, 1)),$$

find $P_{C \leftarrow B}$.

2. Convert $[\mathbf{x}]_B = (3, 2)^T$ to C -coordinates and explain the purpose of the transition matrix in one sentence.

3. For

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

compute $P^{-1}AP$.

4. Verify that A and $P^{-1}AP$ have the same trace and determinant.

5. Use the diagonal representation from Problem 3 to compute A^5 .

6. Explain why the order of vectors in a basis matters for coordinate vectors and transition matrices.

Section 6.5 - Applications of Linear Transformations

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Identify the transformations represented by

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

2. Reflect the point $(3, -2)$ across the line $y = x$.
3. Apply the horizontal shear $H(1/2)$ to the triangle with vertices $(0, 0)$, $(4, 0)$, and $(0, 2)$. Compare the original and transformed areas.
4. Rotate $(2, 0)$ counterclockwise by 30° .
5. Find the matrix for a horizontal stretch by 2 followed by a 90° counterclockwise rotation. Explain why the rotation matrix appears on the left in the product.
6. Rotate $(1, 2, 3)$ counterclockwise by 90° about the z -axis. Explain why the z coordinate is unchanged and state how the right-hand rule determines positive rotation.

7. Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

Find $\det(A)$ and state exactly how A scales planar area. Apply the result to a region of area 5.